

REGULATORY COMPLIANCE FRAMEWORK Sub-Headline: URMOSSES Alignment with U.S. Federal and State Mandates

1. Accessibility & Inclusion (ADA / Section 508)

- **Requirement:** Ensuring full access for the 62+ demographic.
- **Compliance:** The URMOSSES interface and vehicle physical design strictly adhere to the **Americans with Disabilities Act (ADA)** and **Section 508** standards.
- **Features:** High-contrast visual modes, screen-reader compatibility (Text-to-Speech), and physical low-step entry for mobility-impaired passengers.

2. Automated Driving Safety (NHTSA / SAE)

- **Classification:** Operating under **SAE Level 4** (High Automation) within defined Operational Design Domains (ODD).
- **Safety Mandates:** Full integration of **NHTSA** guidance for automated driving systems, including **SOTA** (Safety Over-the-Air) updates and rigorous fail-safe protocols.
- **L1_Safety_Core:** Implementation of a deterministic safety layer that ensures collision avoidance independent of cloud connectivity.

3. Infrastructure Interoperability (MDS / V2X)

- **MDS 2.0+ Compliance:** URMOSSES is engineered to communicate via the **Mobility Data Specification (MDS)** Agency & Provider APIs.
- **V2X Hub:** Hardware-ready for **C-V2X (Cellular Vehicle-to-Everything)** to ensure real-time interaction with LA's Smart City infrastructure.

LIABILITY MITIGATION & DATA GOVERNANCE

1. Data Privacy (CCPA / CPRA)

- **Framework:** Full compliance with the **California Consumer Privacy Act (CCPA)** and its amendment (CPRA).
- **Sovereign_Connector:** Our Mature Data Strategy includes a "Privacy-by-Design" approach.
- **User Control:** Built-in "Opt-Out" mechanisms for data monetization and granular consent management for all personal data (PII).

2. Forensic Traceability (The AUDIT_LOG)

- **The "Black Box" Principle:** Every system decision is recorded in an immutable **AUDIT_LOG**.
- **Liability Support:** Provides verifiable, timestamped data for insurance providers and regulatory audits in the event of an incident.
- **Integrity:** The log is cryptographically secured to prevent tampering or unauthorized access.

3. Cyber Security (ISO 21434)

- **Standard:** Adherence to **ISO/SAE 21434** for road vehicle cybersecurity.
- **Protection:** **CYBER_EC** module provides cryptographic protection of the foundation layer against external manipulation or unauthorized remote access.

4. Operational Risk Mitigation

- **Geofencing:** Strict ODD enforcement via GPS and semantic modeling to prevent unauthorized area operation.
- **Remote Management:** Real-time fleet monitoring and emergency override capabilities for authorized LADOT operators.